

STR-MON Test Jig

Tester User Guide - production test procedure for STR-MON PCB assemblies

Document type: Operator procedure

Firmware: 0.7.0 or later

Serial terminal: Optional only

Normal result method: Jig LEDs and buzzer

Important: The PCB fitted into the jig is the DUT. The serial monitor is not required for normal testing. It is only used for higher-level debug or engineering checks.

1. Jig Overview

The STR-MON test jig checks the DUT using the jig contacts, clamp, TEST button, indicator LEDs, and automated firmware test sequence.

Photo 1 - Jig Overview

Insert overview photo showing DUT fitted in the jig and clamp position.

Photo 2 - DUT Clamped

Insert side/top photo showing the clamp pressing the DUT into the test position.

Controls And Indicators

Item	Meaning	Tester action
Ready LED - Green	Slow flashing means no DUT. Solid means DUT detected and ready to test.	Fit DUT until Ready becomes solid.
Pass LED - Red	Solid after a successful test run.	Record/pass the DUT and remove it from the jig.
Fail LED - Yellow	Solid after a failed test run.	Remove DUT and place it in the fail/rework area.
TEST button	Starts the automated test when DUT is ready.	Press once after Ready LED is solid.
Buzzer	Short beeps confirm DUT detected, test start, and final result.	Use with LEDs as operator feedback.

Photo 3 - LED And Button Close-Up

Insert close-up showing Ready, Pass, Fail LEDs and TEST button.

Photo 4 - Rear / Connector View

Insert photo showing jig connector area and DUT alignment features.

2. Normal Test Procedure

1. Power the test jig.
2. Confirm the Ready LED is slowly flashing. This means no DUT is detected.
3. Place the STR-MON PCB into the jig pocket. Make sure it sits flat and is aligned with the locating features.
4. Close the clamp so the DUT is held firmly against the test contacts.
5. Confirm the Ready LED changes to solid on. The jig should give one short beep.
6. Press the TEST button once.
7. During testing, the Pass and Fail LEDs alternate. Do not remove or touch the DUT while testing is running.
8. Wait for the final LED result and buzzer indication.
9. Remove the DUT and sort it according to the result.

3. Result Interpretation

Jig indication	Meaning	Required action
Pass LED solid	DUT passed the automated test sequence.	Mark or move the DUT as passed.
Fail LED solid	DUT failed one or more automated checks.	Move the DUT to the fail/rework area. Do not retest repeatedly unless instructed.
Ready LED slow flashing	No DUT detected.	Fit a DUT before pressing TEST.
Ready LED solid	DUT detected and ready.	Press TEST once to start.
Pass and Fail LEDs alternating	Test is running.	Wait for the final result.

4. Optional Retest Without Removing DUT

In special cases, the tester may prepare the jig for a second test without removing the DUT.

1. Wait until the DUT has reached Pass or Fail.
2. Release the TEST button if it is pressed.
3. Hold the TEST button for approximately 1.5 seconds.
4. Confirm the jig returns to Ready with the Ready LED solid.
5. Press TEST once to run the automated test again.

Note: This retest function is intended for special cases only. Normal use is to remove the DUT after the first final result.

5. Test Sequence

The operator does not need to follow these steps manually; the jig performs them automatically after the TEST button is pressed.

Order	Automated check	Purpose
1	Open Circuit Test	Confirms the DUT fault relay responds correctly to an open-circuit simulation.
2	Short Circuit Test	Confirms the DUT fault relay responds correctly to a short-circuit simulation.
3	Alarm Positive Test	Checks alarm positive output current behavior.
4	Alarm Negative Test	Checks alarm negative output current behavior.
5	Fault Relay Test	Confirms the fault relay contacts are in a valid complementary state.

6. Tester Checklist

	Check	Accept condition
[]	Jig powered	Ready LED slowly flashes with no DUT fitted.
[]	DUT fitted and clamped	Ready LED solid and one short beep.
[]	TEST pressed once	Pass and Fail LEDs alternate while testing.
[]	Final result observed	Pass LED solid or Fail LED solid.
[]	DUT sorted	Passed DUTs separated from failed/rework DUTs.

7. If Something Looks Wrong

Symptom	Likely cause	Action
Ready LED does not go solid with DUT fitted	DUT not seated, clamp not closed, dirty contacts, or no DUT detection.	Remove DUT, inspect seating/contact area, refit, and clamp again.
Fail LED turns on	DUT failed automated check.	Move DUT to fail/rework area. Do not adjust the jig unless multiple known-good DUTs fail.
Jig resets or LEDs restart during insertion/removal	Possible power dip or relay/contact chatter.	Report to engineering or maintenance.
Serial output is not visible	Serial terminal not open or USB not connected.	No action required for normal testing.

8. Serial Output

The serial terminal is optional. It is intended for engineering debug and high-level test checks only. General testers should use the LEDs and buzzer as the official operating indication.

Normal use: Fit DUT, clamp DUT, press TEST, observe Pass or Fail LED.

Document generated from STR-MON Test Jig firmware release 0.7.0. Update this guide if LED meanings, test order, clamp design, or operator workflow changes.